Lab 4 – OpenPGP

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Client Used: Thunderbird on macOS
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What I Did

In this lab, I learned how to use OpenPGP to protect email messages through digital signatures and encryption.
I created my own OpenPGP key pair in Thunderbird, backed up the secret key, and saved the revocation certificate in a secure location.

After creating the key, I published my public key to both keyservers keys.openpgp.org and keyserver.ubuntu.com.
My **Key ID** is 0xE0390662A1CFFE5B, and the **fingerprint** is 8346 8131 FCE5 F187 1A23 3719 E039 0662 A1CF FE5B.

Next, I downloaded and imported the TA's public key (b544ta@gmail.com) from the OpenPGP keyserver and verified that the fingerprint matched the one provided in the lab instructions.
After verifying, I used my private key to sign the TA's public key to show that I trusted it.

Finally, I sent three separate emails to the TA as required:

1. **Signed only** — Subject: lab4-sd-PGP-snoorzai
   The message was digitally signed but not encrypted.
2. **Signed and Encrypted** — Subject: lab4-se-PGP-snoorzai
   The message was both digitally signed and encrypted.
3. **Encrypted only** — Subject: lab4-e-PGP-snoorzai
   The message was encrypted but not signed.

Each email was successfully sent and appeared in my "Sent" folder with the correct PGP icons.

What I Learned

This lab helped me understand how encryption and digital signatures work together to protect the confidentiality and authenticity of emails.
I also learned how to manage keys safely and how keyservers make it easier for others to verify your identity and exchange secure messages.

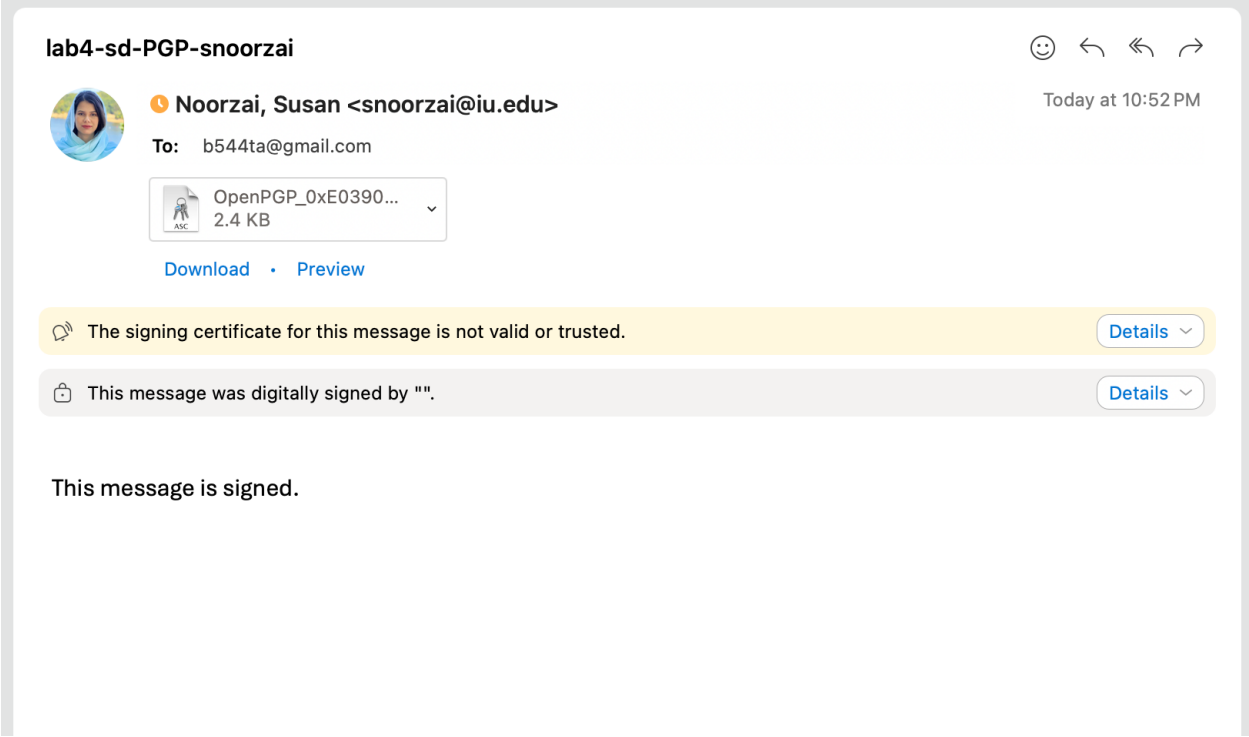
Questions and Answers

1. Can you send signed or encrypted email using webmail such as Gmail? Why or why not?
No, Gmail's normal web interface doesn't support OpenPGP directly.
To use encryption or digital signatures in Gmail, you would need a browser extension such as Mailvelope or FlowCrypt, which performs encryption and signing locally.
The main challenges are securely storing the private key inside the browser, trusting the extension, and dealing with compatibility issues between Gmail's interface and OpenPGP messages.

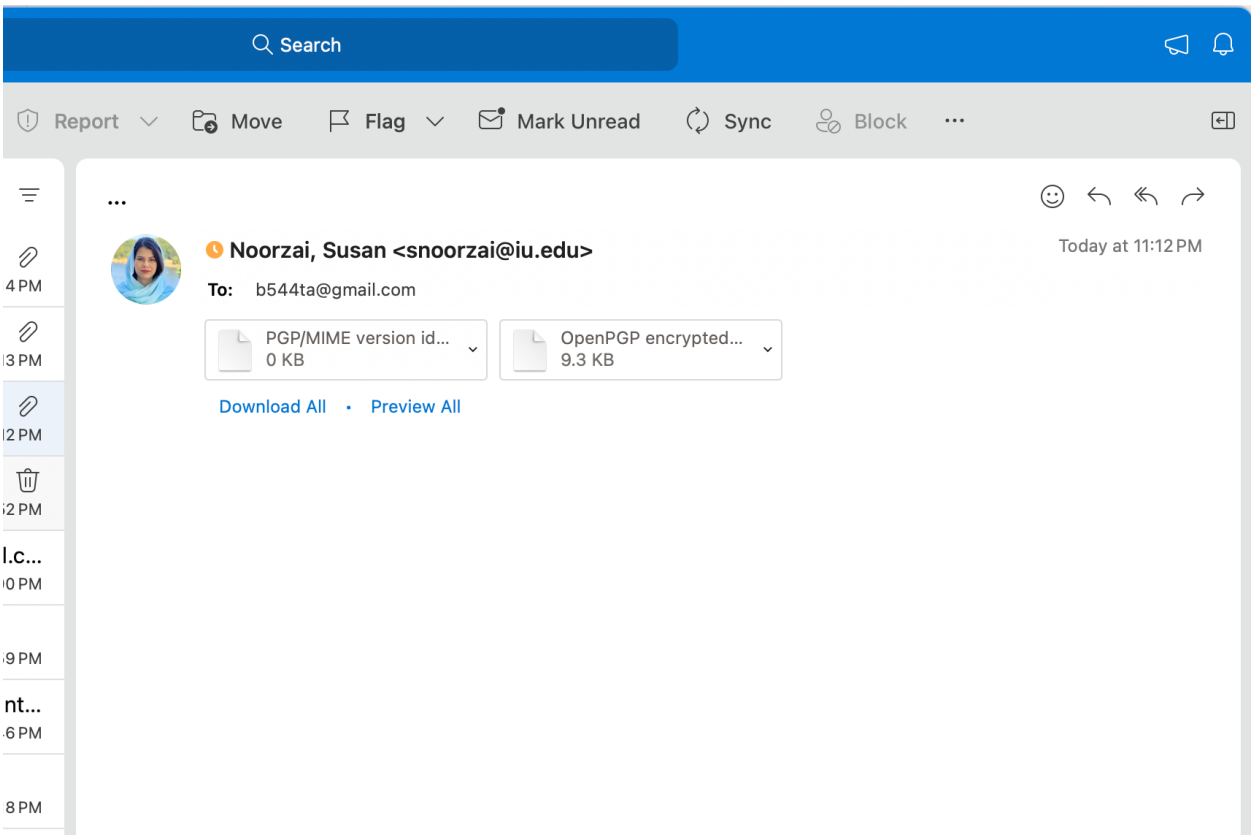
2. Given the fingerprint of a key, why are you convinced that the key file you downloaded hasn't been tampered with?
Because the fingerprint is a unique cryptographic digest of the key.
If the downloaded key had been modified in any way, its fingerprint would be completely different, so matching fingerprints prove the key's integrity.

Attachments / Evidence

- Screenshot 1 – Signed message (lab4-sd-PGP-snoorzai)



- Screenshot 2 – Signed and Encrypted message (lab4-se-PGP-snoorzai)



- Screenshot 3 – Encrypted only message (lab4-e-PGP-snoorzai)

